1. Core loop

The player travels through procedurally generated rooms to defeat enemies and bosses, earning loot, currency, and weapon mastery experience to upgrade their equipment and abilities, allowing them to survive increasingly difficult arenas and repeat the process until defeating the final boss.

2. Mechanics

a. ***<u>Anthony</u>***

i. **Branched world generation**: The world will be procedurally generated consisting of 5 total arenas and 4 special rooms. These rooms are arenas, rest sites, shops, and mystery rooms. The rooms will alternate from special rooms to combat rooms. Out of the three special rooms, only two are spawning giving each a 66% chance of appearing after combat. To select the room you want to go to, the player must walk through one of the doorways that appears after a combat ends. The doors close after the player walks through and gives them the option only to progress to a combat room after.

ii. **Purchasing/Selling**: The player can exchange their currency to buy items to add to their inventory. They can also exchange their items to get currency (the shop will give the player 50% of the items brand new value). The player will use their mouse to look at each item in their inventory and the shop and make selections by left clicking. There will be buttons present that lets the player choose to buy, sell or leave. The cost of items scale with its rarity. Each tier of rarity increases its cost by an additional 25% from the base (ex: common 100, uncommon 125, rare 150, epic, 175, legendary 200)

iii. **Resting**: At resting site rooms there will be a sleeping bag, a campfire, and a workbench. The player will have to choose one of the three to walk up to and press E to sleep at the sleeping bag and heal for 50% of their maximum health, upgrade one of their stats by 20% (ex: increase speed by 20%), or upgrade an eligible item's mastery progression by 20XP

iv. **Mystery Rooms**: Mystery rooms are activated as soon as the player enters them and they shuffle through multiple different things (each with a 20% chance of happening):

- Another arena fighting room
- A curse/debuff bestowed upon you (ex: -10% to a random stat you have)
- Robbery: You lose 25% of your currency or a random item in your inventory
- A buff bestowed upon you (ex: +10% to a random stat you have)
- Finding a skeleton with a random item for you to take.

b. ***<u>Jadin</u>***

i. RNG / Item Drop System

- **Input:** No direct input is required. Item drops are automatically generated when an enemy is defeated.
- **Rules and Numbers:**
- When an enemy is defeated, it has a chance to drop an item that the player can collect. The item type and rarity are randomly determined using a probability-based loot system. Different enemy types have different drop chances and available loot pools.
- Regular enemies have a 60% chance to drop an item.
- Minibosses have a 100% chance to drop at least one item.
- Bosses have a 100% chance to drop at least one item, including a guaranteed rare or higher rarity item.
- When an item drops, its rarity is randomly determined using the following probabilities:
    - Common: 59%
    - Uncommon: 25%
    - Rare: 12%
    - Epic: 3%
    - Legendary: 1%
- Bosses use a separate guaranteed rare-or-higher drop rule instead of the regular rarity distribution.
- Items can include weapons, consumables, and upgrade materials. Higher-rarity items can have stronger effects, increased damage, or additional benefits.
- **How It Interacts With My Other Mechanic:**
    - The RNG Item Drop System affects the Inventory System because the type and rarity of the generated drops determine what items the player can collect, store, equip, and use.

ii. **Inventory System**

- **Input:**
    - E – Pick up an item when standing within 2 meters of it.
    - Tab – Open or close the inventory.
    - Left Click – Select an item in the inventory.
    - Left Click on Equip/Use – Equip a weapon or use a consumable.
    - Left Click on Drop – Remove an item from the inventory and place it back into the world.
- **Rules and Numbers:**
- The player has an inventory with 12 available item slots. Each weapon occupies one slot and cannot be stacked. Consumable items and upgrade materials can stack up to 5 of the same item in one slot.
- When an enemy drops an item, the player must walk within 2 meters of it and press E to collect it. The item is automatically placed into the first available inventory slot.

- If the inventory is full, the player cannot collect additional items until they make room by using, selling, or dropping something.
- The player can equip one weapon at a time from their inventory and switch between collected weapons. Consumable items can also be used directly from the inventory.
- **How It Interacts With My Other Mechanic:**
- The Inventory System affects the RNG Item Drop System because the player's available inventory space determines whether newly generated drops can be collected. If the inventory is full, the player must make room before collecting additional drops.

c. ***Evan***

i. **Enemies - (regular enemies, bosses, minibosses, final boss)**

- Its name, and the input that does it.
  - Enemies, if the player is visible or not, how it is attacked and how it attacks, what arena it is in from the arena mechanic.
- The rules and the numbers.
  - Each enemy has 2 attacks and a set amount of health. It can only spawn in certain rooms and once health is 0, it must disappear.

ii. **Arena - (3 types of arenas with different mechanics and environments, boss room)**

- Its name, and the input that does it.
  - Arena takes in what is chosen from the branched world generation mechanic.
- The rules and the numbers.
  - Each room has an individual mechanic whether it be lava or acid rooms with subtle stage hazards with enemies roaming. The stage determines what enemies can spawn in.

d. **Michael**

i. **Combat System**

- **Inputs:**
  - Left Click – Light Attack
  - Right Click – Heavy Attack
  - Space – Dodge
  - F – Parry
- **Weapons:**
  - **Daggers:** Fast, short range. 8–15 damage.
  - **Swords:** Balanced speed and range. 10–20 damage.
  - **Greatswords:** Slow, high damage. 18–30 damage.
  - **Spears:** Long range, moderate damage. 10–18 damage.
- **Rules and Numbers:**
  - Players have 100 stamina, regenerating 15 per second.
  - Attacks consume 5–30 stamina depending on the weapon.

        - Dodging costs 15 stamina and provides 0.3 seconds of invincibility.
        - Parrying costs 10 stamina and staggers enemies for 1.5 seconds.

ii. Weapon Mastery System

- **Input:** Uses combat controls.
- **Rules and Numbers:**
    - Defeating enemies grants 5 mastery XP for the equipped weapon.
    - At 25 XP, players unlock a new combo.
    - At 50 XP, players unlock a special attack costing 30 stamina.
    - Each weapon has its own mastery progression.
- **How They Interact:**
    - Combat allows players to fight using different weapons, while Weapon Mastery rewards continued use by unlocking new attacks and combos.
    - Then, for each of your mechanics, one sentence on which of your others it changes and how.

3. How they plug in.

Owner	Their own mechanics	How they change each other
Anthony	World Generation ↔ Purchasing/Selling	World generation determines when shops appear; shop purchases affect the player's resources and readiness for future branches
Anthony	Resting ↔ Mystery Rooms	Resting can recover stats lost through mystery-room penalties; mystery-room buffs and curses influence which rest upgrades are valuable
Jadin	RNG Drops ↔ Inventory	Drops determine available items; inventory capacity determines which drops can be collected

Evan	Enemies ↔ Arenas	Enemies influence how players navigate hazards; arena hazards and layouts change enemy movement and attacks
Michael	Combat ↔ Weapon Mastery	Combat earns mastery XP; mastery unlocks additional combat attacks and combos

Cross-member interactions:

- Anthony → Jadin: Purchasing and Selling changes the inventory's contents and available space.
- Jadin → Michael: Inventory determines which weapons can be equipped during combat.
- Evan → Michael: Enemy attack patterns, health, and damage values change combat strategy.
- Michael → Evan: Combat attacks, dodges, parries, and mastery abilities change how enemies are defeated.

Anthony's four mechanics also need to affect one another beyond these pairs. For example, World Generation determines access to rest and mystery rooms, while mystery-room rewards and resting upgrades affect the player's readiness and choice of future branches.

4. Win and Lose Conditions

Win: The player successfully progresses through the generated combat arenas, defeats the required bosses, and defeats the final boss.

Lose: The player's health reaches zero from enemy attacks or environmental hazards, ending the current run.

For the initial version, losing restarts the run from the beginning.

5. **Art direction.** Your art sheet: palette as hex values, style suffix, avoid list, and anchor asset.

5. Art Direction

Visual Style:

Our game will feature a dark, low-poly, retro horror aesthetic inspired by the game *Inscryption*. The environments will use pixelated textures, muted colors, dramatic lighting, and unsettling atmospheres. The goal is to create a mysterious dungeon-crawling experience where each arena feels dangerous and unpredictable. The game will maintain a consistent visual style across combat arenas, shops, rest areas, mystery rooms, and boss encounters.

Color Palette (Hex Values):

- Near Black: #171711 – Shadows, dark environments, and backgrounds.

- Moss Green: #353D2F – Environmental details, vegetation, and eerie lighting.

- Wood Brown: #66533D – Wooden structures, furniture, and dungeon decorations.

- Candle Gold: #C2A46B – Torches, lighting, and interactive objects.

- Blood Red: #8C302B – Enemy attacks, damage indicators, and dangerous areas.

Style Suffix:

Low-poly PS1-era horror, pixelated textures, dark fantasy dungeon, muted earthy colors, atmospheric fog, harsh shadows, warm candlelight, unsettling environments, minimal detail, retro 3D graphics, Inscryption-inspired atmosphere.

Avoid List:

- Bright, saturated cartoon colors.

- Smooth, realistic AAA graphics.

- Anime-style characters.

- Clean futuristic or sci-fi environments.

- Excessively detailed models.

- Modern glossy materials.

- Overly bright environments.

- Realistic human proportions and facial features.

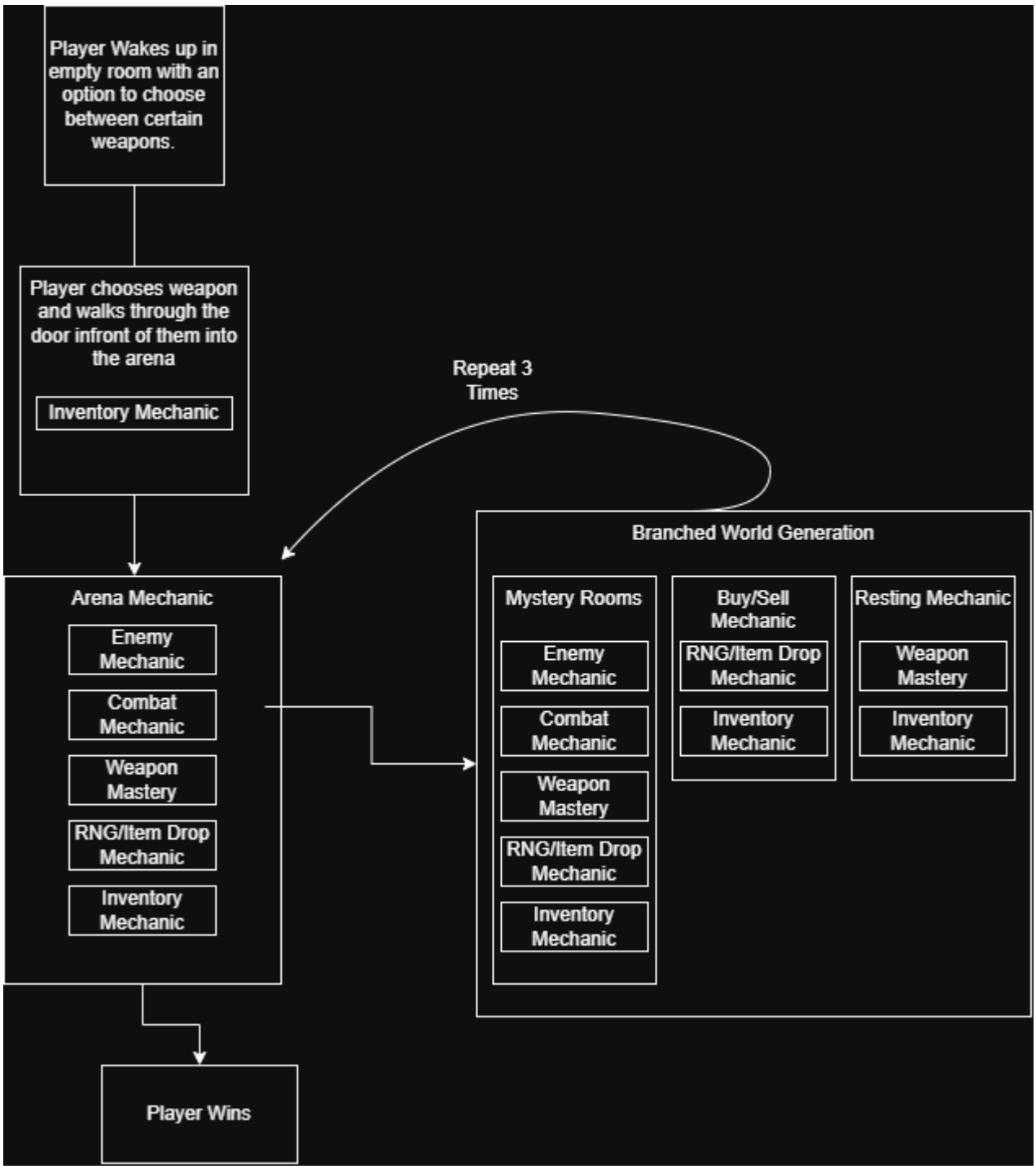
Anchor Asset:

Our anchor asset will be a low-poly dungeon combat arena featuring rough wooden structures, cracked stone floors, flickering torches, heavy shadows, and an ominous doorway leading to the next room. This asset will establish the overall visual style for the game and serve as a reference when creating additional environments, enemies, weapons, and interactive objects.

Although the visual style is inspired by *Inscryption*, our game will focus on real-time combat, enemy encounters, weapon mastery, and procedurally generated arenas rather than card-based gameplay.

6. Architecture.

A diagram, drawn before you build: your scenes and scripts, where each mechanic's state lives, and which script reads which. Boxes and arrows are enough: one box per scene or script, labeled with the mechanic state it holds, and an arrow from each script to every script it reads. Your code will differ from it, and that costs nothing here.



7. Scope and Cuts

Development Scope

Our team plans to have all ten mechanics implemented and functional by **November 18, 2026**. Between November 18 and December 2, we will focus on bug fixes, balancing, polishing, and ensuring that all mechanics work together properly.

Mechanic	Owner	Completion Date
Branched World Generation	Anthony	November 18
Purchasing/Selling	Anthony	November 18
Resting	Anthony	November 18
Mystery Rooms	Anthony	November 18
RNG Item Drop System	Jadin	November 18
Inventory System	Jadin	November 18
Enemy System	Evan	November 18
Arena System	Evan	November 18
Combat System	Michael	November 18
Weapon Mastery System	Michael	November 18

November 18 – Initial Completion

By November 18, every team member will have implemented their assigned mechanics. All core systems should be functional, allowing the player to navigate rooms, fight enemies, collect items, manage inventory, purchase upgrades, and progress through the game.

December 2 – Final Polish

Between November 18 and December 2, our team will focus on testing and improving the existing mechanics rather than introducing new features. This includes fixing bugs, balancing enemies and weapons, improving animations and visual effects, refining the user interface, and ensuring that all systems interact correctly.

Cuts

None. Our team has not dropped any mechanics from the original kickoff pitches. All ten proposed mechanics are currently included in the development scope. If any mechanics are removed later due to time constraints or technical difficulties, we will document those changes and explain the reasons for removing them.